1. DATABRICKS

HIVE METASTORE

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HIVE METASTORE

- Hive metastore is a catalog that stores metadata i.e. data about data.
- It is used to support SQL.

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How does spark locate the data?

- Hive metastore acts like a RDBMS.
- Any RDBMS like sql server or oracle store data in files which is the underlying for all tables, same concept is applied in hive metastore which enables tables to locate the files which have the data stored.
- Hive metastore has been since the Hadoop ecosystem.

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Catalog

- Hive metastore is a catalog that stores metadata.
- This catalog stores data about all the tables and its underlying files which store the data.

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SQL Support

- Data is stored in raw format in data lake.
- It is stored in different file formats like csv, json, txt, etc.
- Hive metastore enables us to directly query these files and perform operations with SQL.

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One to one relationship

- Important, every Databricks workspace has its own metastore.
- Hence, if you have multiple Databricks workspaces they will individually have their own hive metastore.

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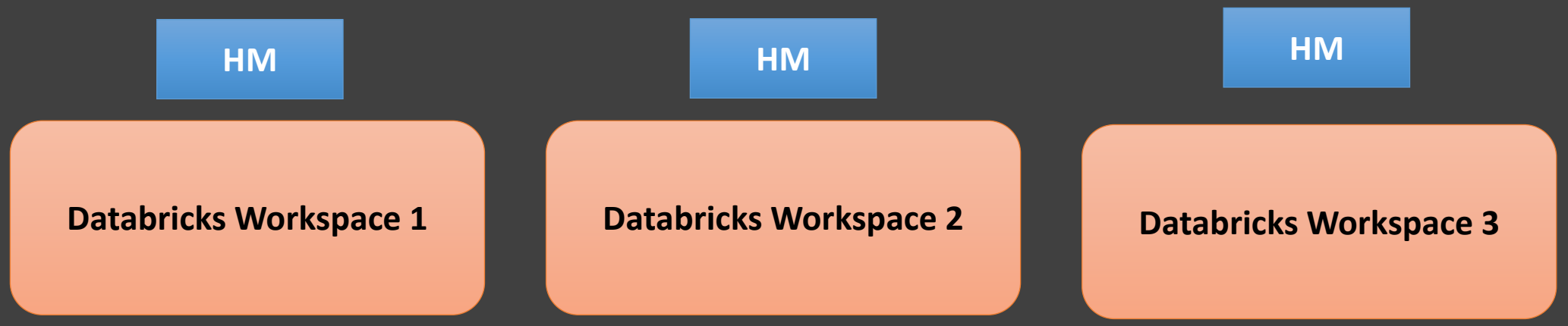
Issue with existing Hive catalog



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Issue with existing Hive catalog

- As the number of Databricks workspaces increase, it becomes difficult to manage data assets.
- It brings up challenges in data governance, security and access management.

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Solution

- Unity Catalog!
- It combines all metastores under one umbrella.
- Enables data governance, security roles and access management ease.

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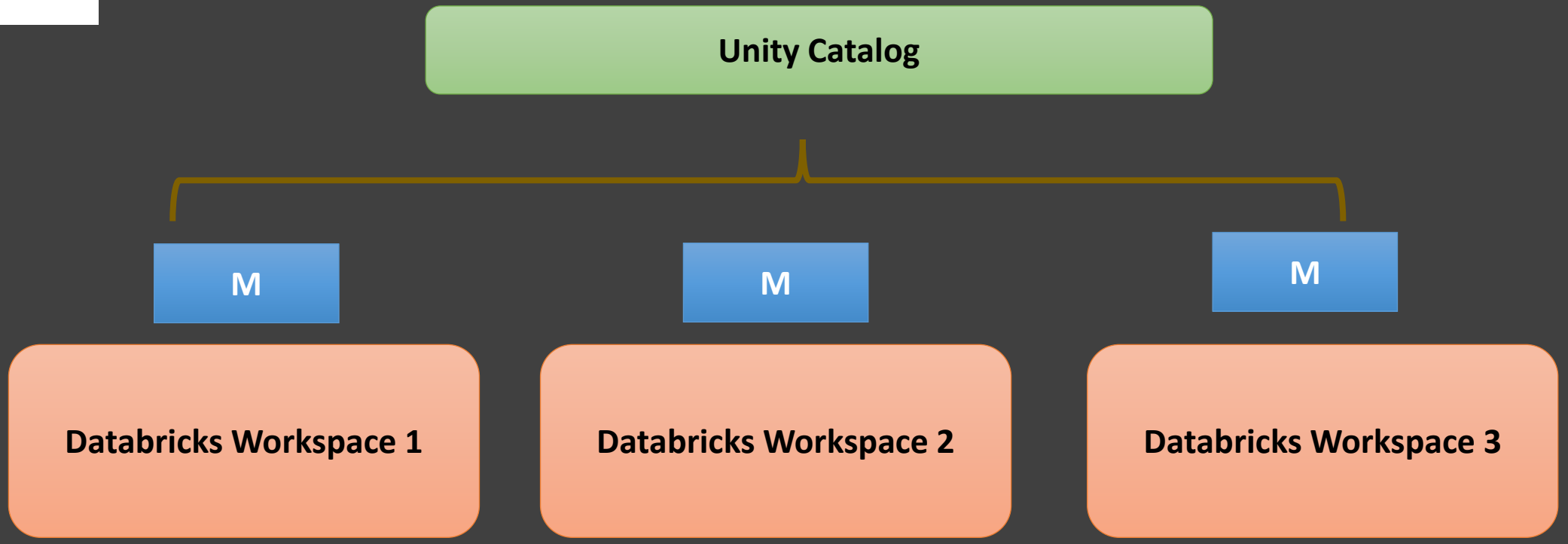
Issue with existing Hive catalog



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2. DATABRICKS

INTRODUCTION TO UNITY CATALOG

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Unity Catalog

- Developed by Databricks, Unity Catalog is the only unified and governance solution.
- Its is built for Data and AI assets.
- Unity catalog enables organizations seamlessly govern both structured and unstructured data , be it in any format.
- It enables managing objects like ML models, notebooks, files and other objects across any cloud or platform.

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Unified Data Access Control

- As we deal with data, mostly stored in form of tables, it is necessary to grant access to the right individuals who only need it.

- This requirement calls for managing of permissions on tables or objects.

- Different level of permissions can be implemented like read only, write only to specific user or group of users with added row/column level permissions.

- Hence, Unity Catalog becomes a centralized place to
- administer data, manage permissions and audit.

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Requirements

- Unity catalog is only available in the premium pricing tier.
- It is available across all leading cloud platforms Azure, GCP and AWS.
- There is also an open source version of unity catalog.

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Unified Catalog

- As one of the benefits of Unity catalog is acts as an umbrella for all metastores, additionally it can store all Databricks workspace objects like tables, ML models, notebooks, etc under it.

- Every single thing stored inside Unity Catalog is termed as an object.

- These objects can be granted permissions to access.

- It is also possible to blend data from other catalogs like
- existing hive metastore.

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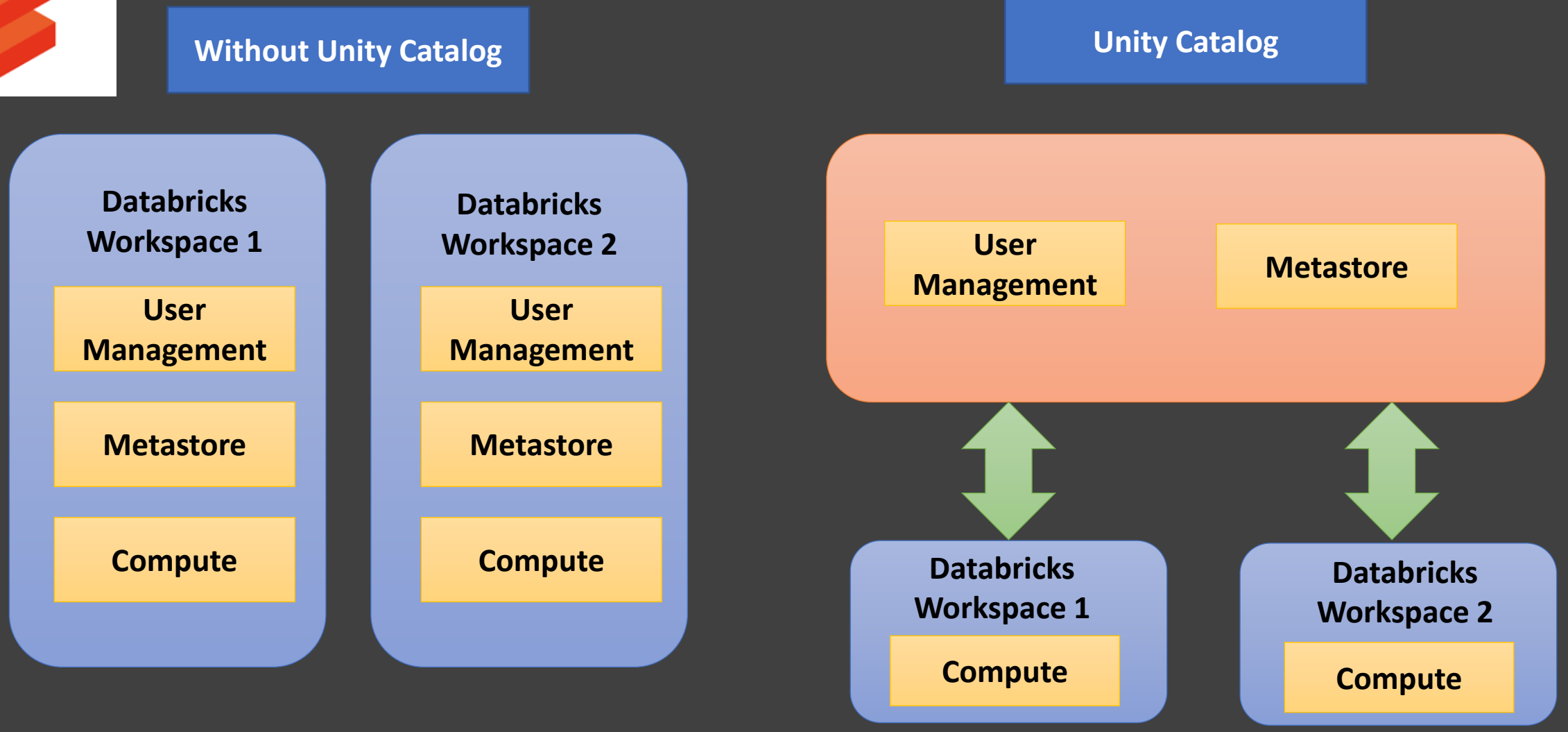
Unity Catalog



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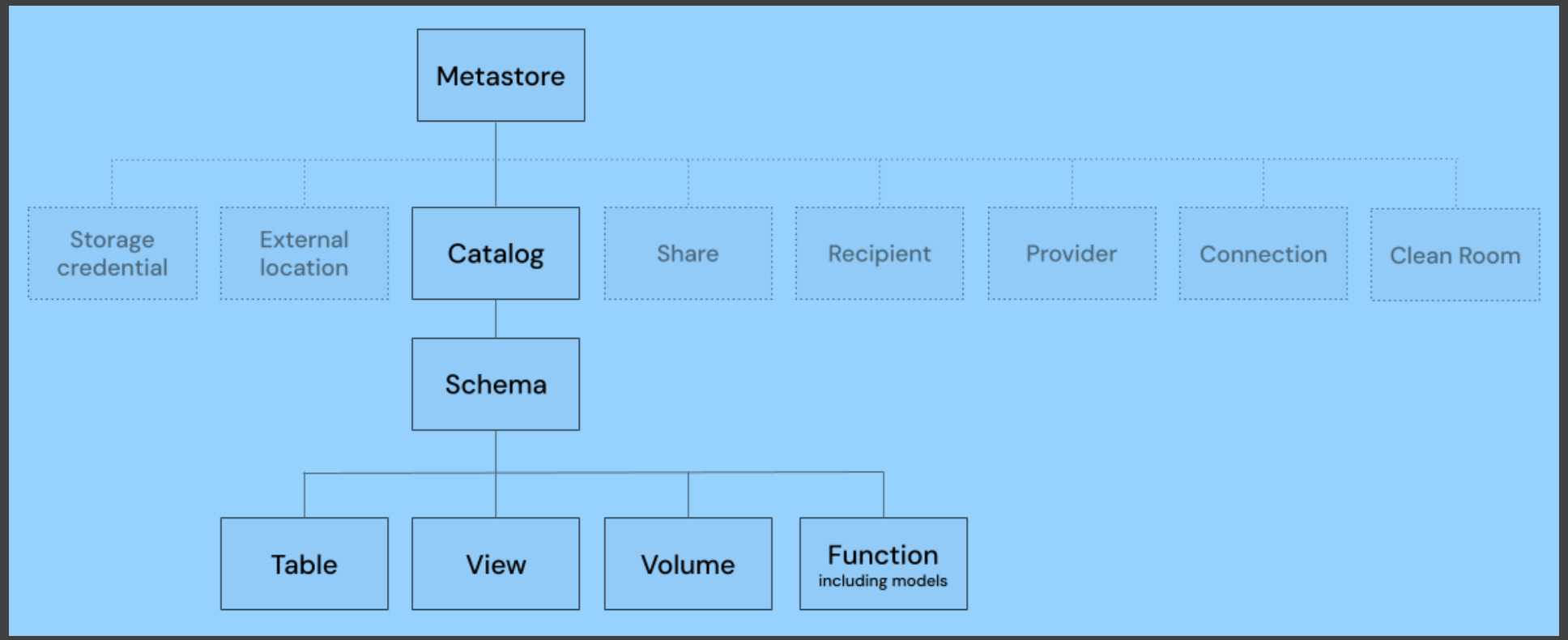
Unity Catalog Object Model



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Unity Catalog

- In a unity catalog all metadata is stored in the metastore.
- You have to create the metastore manually.
- The hierarchy of objects is divided into 3 levels under metastore, which is 3 level namespace.
- Catalog.schema.table

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Metastore

- Metastore is the top-level container for metadata in Unity Catalog.
- It registers data about data and AI assets and permissions that govern them.
- For a Databricks workspace to use Unity Catalog, it must have a Unity Catalog metastore attached to it.
- You should have one metastore for each region in which you have Databricks workspaces.

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Metastore

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6. DATABRICKS

DATA GOVERNANCE

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Data Governance

- Data governance is a strategic framework designed to manage and control an organization’s data assets throughout their lifecycle.

- It ensures data is accurate, secure, and used effectively across the business.

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Data Governance Benefits

- **Structured Framework**: Establishes principles, policies, and standards to manage data consistently.
- **Lifecycle Management**: Covers data from creation and storage to usage and disposal.
- **Alignment with Business Goals**: Connects data practices with organizational strategy for better outcomes.

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Data Governance Benefits

- **Improved Data Quality**: Ensures data is accurate, consistent, and reliable.
- **Enhanced Data Visibility**: Provides transparency into data sources, ownership, and usage.
- **Security and Access Control**: Protects data from unauthorized access and misuse.

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Data Governance Benefits

- **Regulatory Compliance**: Helps meet regulatory requirements such as HIPAA, FedRAMP, GDPR, CCPA. This protects organization's reputation and builds stakeholder trust.

- **Supports Decision-Making**: Makes trustworthy data easily accessible for informed business decisions.

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Key Elements of Data Governance

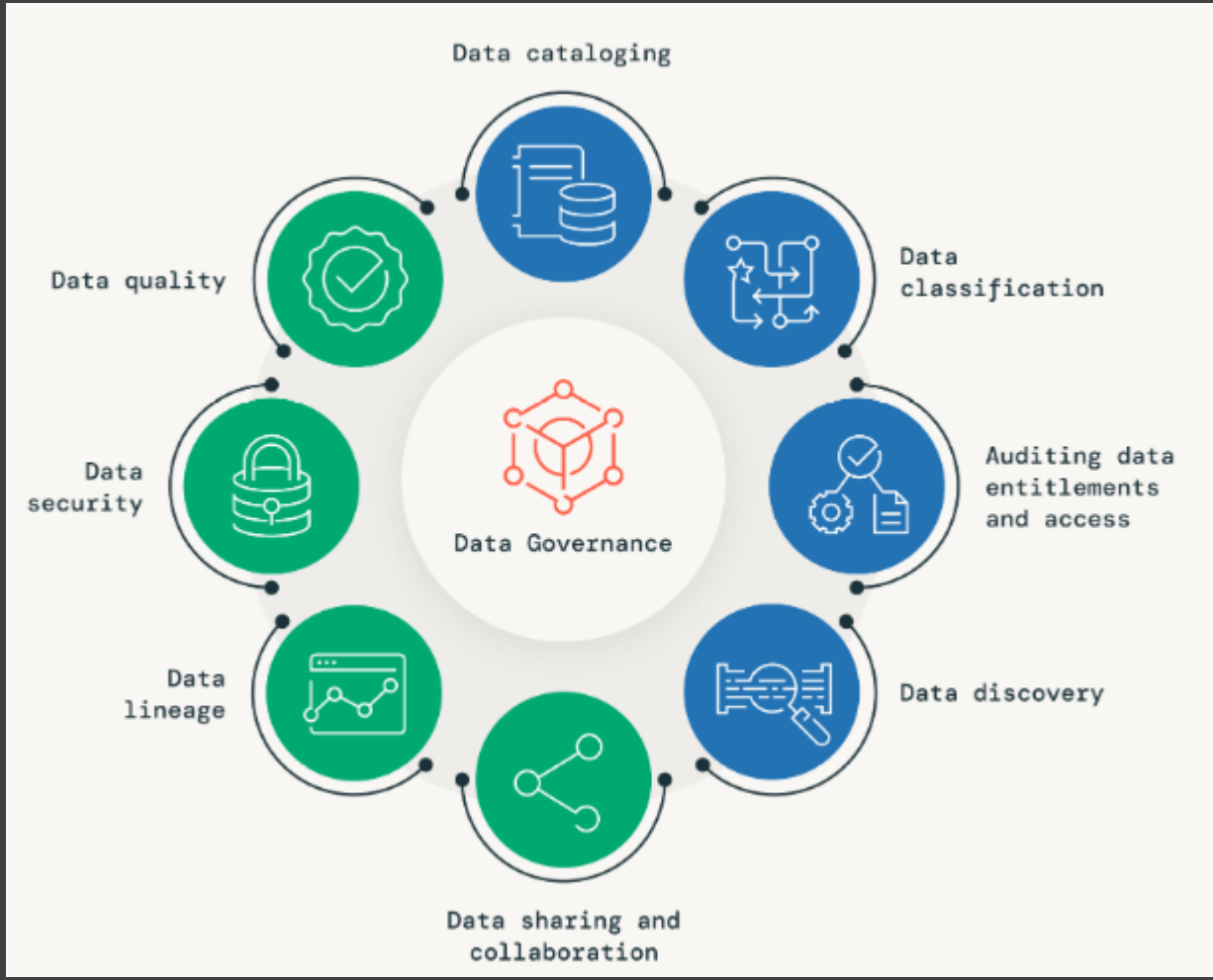

Image source: Databricks

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Data Cataloging

- A data catalog is a centralized metadata repository that helps stakeholders easily discover, understand, and access organizational data. It enhances governance, collaboration, and access control by providing searchable insights into data structure, usage, and location.

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Data Quality

- High data quality is essential for trustworthy analytics and effective decision-making, making it a core pillar of data governance.

- By ensuring accuracy, completeness, and compliance, organizations can enhance data reliability and drive better business outcomes.

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Data Security

- Data security in governance ensures that sensitive information is protected through strict access controls, allowing only authorized users to access specific data.

- With rising privacy regulations and breach risks, robust access management is key to safeguarding data and maintaining trust.

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Data Lineage

- Data lineage offers a complete view of data flow, helping organizations track data from source to consumption for improved trust and quality.

- It supports compliance, simplifies audits, and enables faster root cause analysis, making it a key element of effective data governance.

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Data Sharing & Collaboration

- Secure data sharing and collaboration are essential for modern businesses, enabling controlled data exchange across teams, partners, and platforms.

- Tools like data cleanrooms and marketplaces ensure privacy, compliance, and innovation, making data sharing a key part of governance.

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Data Discovery

- Data discovery is essential for enabling analytics, AI, and ML by making data assets easily accessible across the organization.
- It supports collaboration, prevents duplication, and accelerates innovation—making it a vital part of any data governance strategy.

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Data Audits & Access

- Data access auditing is vital for identifying who accesses what data, helping reduce risks of misuse and regulatory noncompliance.

- Strong audit practices ensure security, compliance (e.g., GDPR, CCPA), and maintain customer trust in regulated environments.

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Data Roles & Responsibilities

Chief Data Officer

- Chief Data Officer (CDO) is the head of governance team. Ultimately, responsible for data's security, accessibility and usability.
- A CDO's role involves setting the system up, securing funding and staff for its operation.

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Data Roles & Responsibilities

Data Owners

- Data owners are individuals or teams tasked with the technical oversight of data sets.
- They are often responsible for determining which team members can access specific types of information. If poor or missing policies result in a data breach, these individuals may be held accountable.
- Because of the importance and responsibility tied to this role, data owners are usually senior members within the organization.

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Data Roles & Responsibilities

Data Stewards

- To support the daily operations of data governance workflows, data owners and Chief Data Officers (CDOs) often appoint data stewards.
- These stewards are responsible for putting the established governance program into action and ensuring that both existing and newly acquired data is properly managed.
- Their duties include monitoring compliance by both employees and customers, and promptly escalating any issues that may arise.

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Data Roles & Responsibilities

Data Governance Committees

- This committee will serve as the primary authority for developing data-related policies within your organization.
- Typically composed of senior executives and data owners, the committee brings together individuals with a strong focus on data security and usability.
- Once policies are approved, the committee may define procedures for data stewards to implement and also act as the decision-making body for resolving any disputes that arise.

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Data Roles & Responsibilities

Data Consumer

- A data consumer is anyone who uses data to perform their job or make informed decisions. This can include analysts, data scientists, business users, marketers, product managers, and other stakeholders across the organization.

- Each data consumer has a specific level of access based on roles assigned to them.

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